

Lecture: Eigendecomposition and PCA

*Date: October 15th, 2025**Author: Surbhi Goel*

1 Recap and Preview

Last time we added geometric structure through inner products and projections. We saw the solution to linear regression $\hat{\theta} = (X^T X)^{-1} X^T Y$ is an orthogonal projection.

Today: some matrices become diagonal in the right basis. This is eigendecomposition, and it's the key to Principal Component Analysis.

2 Eigenvalues and Eigenvectors

2.1 Definition and Intuition

- Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar $\lambda \in \mathbb{R}$ is an **eigenvalue** and a non-zero vector $v \in \mathbb{R}^n$ is a corresponding **eigenvector** if:

$$Av = \lambda v$$

- Intuition:** Most vectors change direction when transformed by A . But eigenvectors are special—they only get scaled by λ , without changing direction.

- Example:** For $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, we have:

$$- A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \text{eigenvector } \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ with eigenvalue } \lambda_1 = 2$$

$$- A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \text{eigenvector } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \text{ with eigenvalue } \lambda_2 = 3$$

- If v is an eigenvector, so is any scalar multiple cv (for $c \neq 0$):

$$A(cv) = cAv = c\lambda v = \lambda(cv)$$

We typically normalize eigenvectors to have unit length: $\|v\| = 1$.

2.2 Finding Eigenvalues

- From $Av = \lambda v$, we get $(A - \lambda I)v = 0$. For non-trivial solutions, $A - \lambda I$ must be singular.
- Therefore, λ is an eigenvalue if and only if $\det(A - \lambda I) = 0$ (the **characteristic equation**).
- An $n \times n$ matrix has at most n eigenvalues.

2.3 Properties of Eigenvalues

- A and A^T have the same eigenvalues (though possibly different eigenvectors).
- Similar matrices have the same eigenvalues. Recall: two matrices A and $\tilde{A}$ are **similar** if $\tilde{A} = S^{-1}AS$ for some invertible S , meaning they represent the same linear transformation in different bases. This implies:
 - Eigenvalues are properties of the **linear transformation**, not the particular matrix representation
 - Changing basis doesn't change eigenvalues
- For symmetric, positive definite matrices: all eigenvalues are real and positive.
- For symmetric, positive semi-definite matrices: all eigenvalues are real and non-negative.
- **Useful facts:**
 - Sum of eigenvalues = trace: $\sum_{i=1}^n \lambda_i = \text{tr}(A) = \sum_{i=1}^n a_{ii}$
 - Product of eigenvalues = determinant: $\prod_{i=1}^n \lambda_i = \det(A)$

3 Eigendecomposition

3.1 Diagonalization

- A diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ is especially nice: easy to compute powers, inverses, determinants.
- **Eigendecomposition:** If $A = PDP^{-1}$ where D is diagonal, then:
 - The λ_i on the diagonal of D are eigenvalues of A
 - The columns of P are corresponding eigenvectors
- This is exactly the change of basis from Lecture 12! The matrix P changes from the eigenbasis to the standard basis, and in the eigenbasis, A looks like diagonal D .
- **Why?** Write $P = [v_1 | \dots | v_n]$ where $Av_i = \lambda_i v_i$. Then:

$$AP = [Av_1 | \dots | Av_n] = [\lambda_1 v_1 | \dots | \lambda_n v_n] = PD$$

3.2 Spectral Theorem

Symmetric matrices have especially nice eigendecompositions.

Theorem 1 (Spectral Theorem). *If $A \in \mathbb{R}^{n \times n}$ is symmetric, then we can write:*

$$A = Q\Lambda Q^T$$

where Q is orthogonal (columns are orthonormal eigenvectors) and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ contains the real eigenvalues.

- This is similar matrices with $S = Q$: we have $\Lambda = Q^{-1}AQ = Q^T AQ$ since $Q^{-1} = Q^T$ for orthogonal matrices!
- **Interpretation:** $A = Q\Lambda Q^T$ means: (1) Q^T rotates to eigenbasis, (2) Λ scales by eigenvalues, (3) Q rotates back.
- Covariance matrices are symmetric and positive semi-definite, so they always have this decomposition with $\lambda_i \geq 0$.

3.3 Other Decompositions as Basis Change

Eigendecomposition is one of several powerful techniques that finds a new basis to simplify a linear transformation. Other decompositions offer different perspectives by changing basis in specific ways:

- **Singular Value Decomposition (SVD, $A = U\Sigma V^T$):** For any rectangular matrix, this finds two different orthonormal bases (for the domain and codomain, with respect to the standard inner product) that make the transformation diagonal. This is key to PCA's practical implementation.
- **QR Decomposition ($A = QR$):** For any rectangular matrix, this finds an orthonormal basis for the column space (Q) such that the transformation in this new basis is upper-triangular (R). This is useful for efficiently solving linear systems.
- **Cholesky Decomposition ($A = LL^T$):** For symmetric, positive-definite matrices, this can be viewed as finding a new basis that is orthonormal with respect to the inner product defined by A (i.e., $\langle x, y \rangle_A = x^T Ay$). In this new basis, the transformation represented by A becomes the identity.

4 Application: Principal Component Analysis (PCA)

PCA finds a low-dimensional representation of high-dimensional data ($x_1, \dots, x_N \in \mathbb{R}^D$) that captures the most information by aligning the new basis with directions of maximum variance.

4.1 The PCA Algorithm

1. **Center data:** Compute $\tilde{x}_i = x_i - \mu$, where $\mu = \frac{1}{N} \sum x_i$.
2. **Compute covariance matrix:** $\Sigma = \frac{1}{N} \sum_{i=1}^N \tilde{x}_i \tilde{x}_i^T$. This $D \times D$ symmetric, positive semi-definite matrix summarizes the data's variance.
3. **Find principal components:** Eigendecompose the covariance matrix: $\Sigma = Q\Lambda Q^T$. The columns of Q (eigenvectors) are the **principal components**. The diagonal of Λ (eigenvalues) gives the variance captured by each component.
4. **Reduce dimensionality:** Select the k eigenvectors corresponding to the k largest eigenvalues to form a matrix $U_k = [u_1 | \dots | u_k]$.
5. **Project data:** The new k -dimensional representation for each point is $z_i = U_k^T \tilde{x}_i$.

4.2 Why It Works: Eigenvectors Maximize Variance

The goal is to find a direction u (unit vector) that maximizes the variance of the data when projected onto that direction. The projection of a data point $\tilde{x}_i$ onto u is $\tilde{x}_i^T u$, and the variance of these projections is:

$$\text{Var}(u) = \frac{1}{N} \sum_{i=1}^N (\tilde{x}_i^T u)^2 = u^T \left(\frac{1}{N} \sum_{i=1}^N \tilde{x}_i \tilde{x}_i^T \right) u = u^T \Sigma u$$

- Using the spectral decomposition $\Sigma = Q \Lambda Q^T$ (with $\lambda_1 \geq \dots \geq \lambda_D$), we can rewrite the variance. Let $v = Q^T u$. Since Q is orthogonal, $\|v\| = 1$. The variance becomes:

$$u^T \Sigma u = (Qv)^T \Sigma (Qv) = v^T Q^T \Sigma Q v = v^T \Lambda v = \sum_{i=1}^D \lambda_i v_i^2$$

- Since $\sum v_i^2 = 1$, this expression is a convex combination of the eigenvalues and is maximized at λ_1 when $v = (1, 0, \dots, 0)$.
- This corresponds to $u = Qv = u_1$, the first eigenvector of Σ . The second principal component, u_2 , is found by maximizing the same variance objective within the orthogonal complement of $\text{span}(u_1)$. This process continues, with each subsequent component maximizing variance in the subspace orthogonal to all previously found components.

Next lecture: we extend these ideas to infinite-dimensional function spaces, leading to the Representer Theorem.